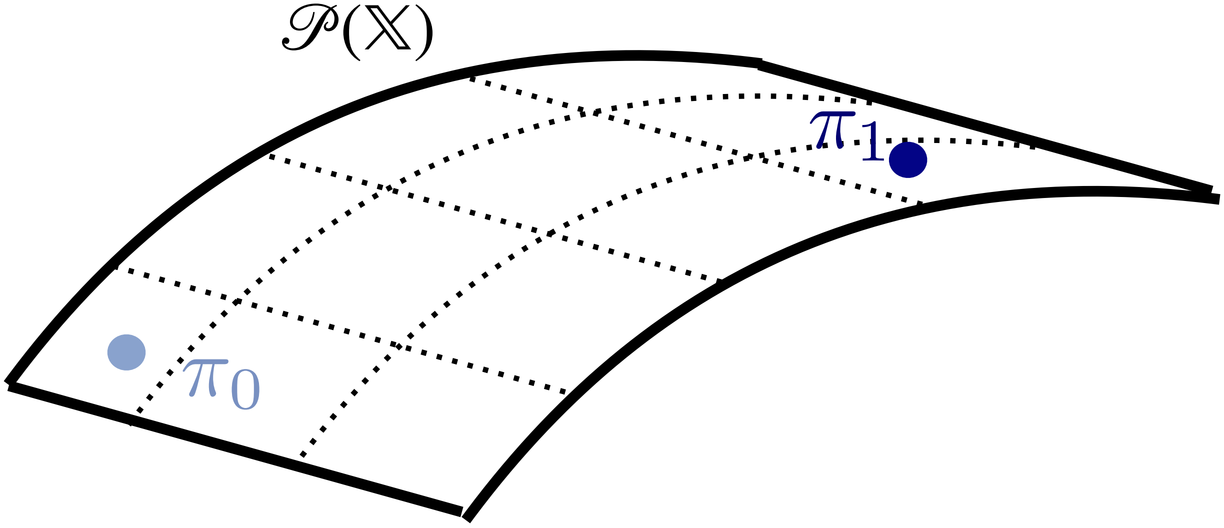
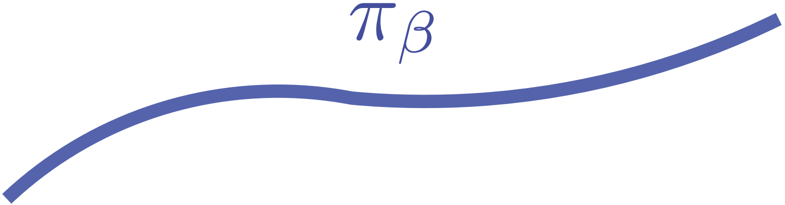
ANNEALING AS A CURVE

- $\beta \mapsto \pi_\beta$ defines a curve in $\mathscr{P}(\mathbb{X})$ parametrised by β with **position** π_β

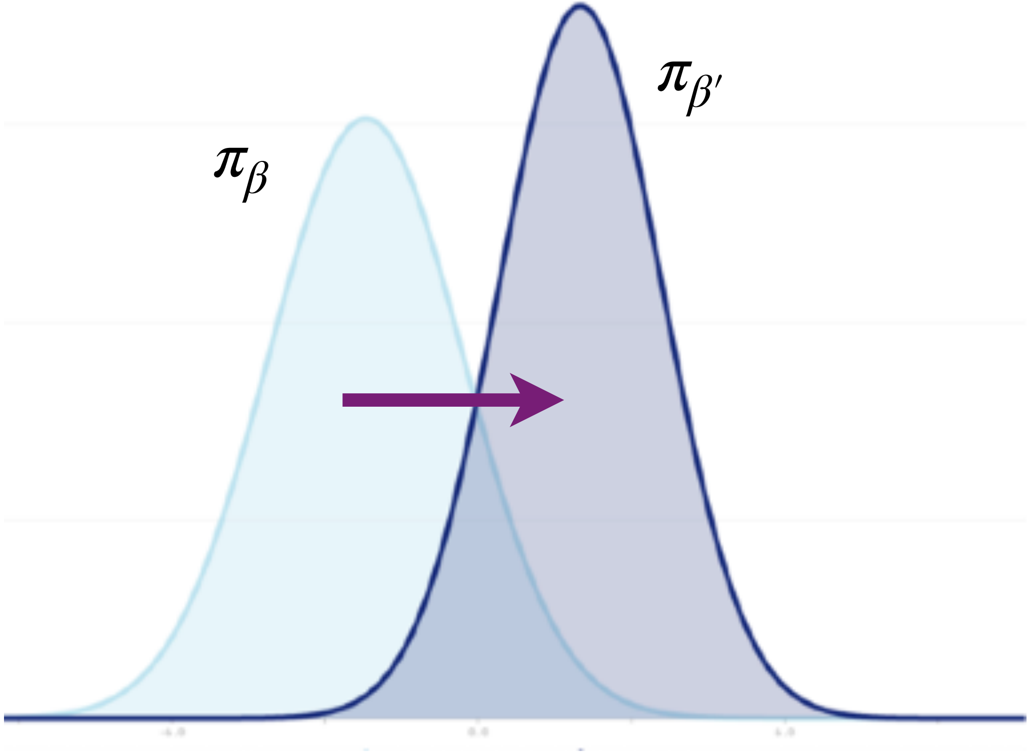

$\mathscr{P}(\mathbb{X})$
π_1
π_0


π_β

- The change in position $\boldsymbol{\Delta\pi}_{\beta,\beta'} : \mathbb{X} \mapsto \mathbb{R}_+$ as the percentage change in likelihood

- Suppose $\mathscr{P}(\mathbb{X})$ is the space of probability densities supported on $\mathbb{X}$

$$\Delta\pi_{\beta,\beta'}(x) = \frac{d\pi_{\beta'}}{d\pi_\beta}(x) - 1 = \frac{\pi_{\beta'}(x) - \pi_\beta(x)}{\pi_\beta(x)}$$

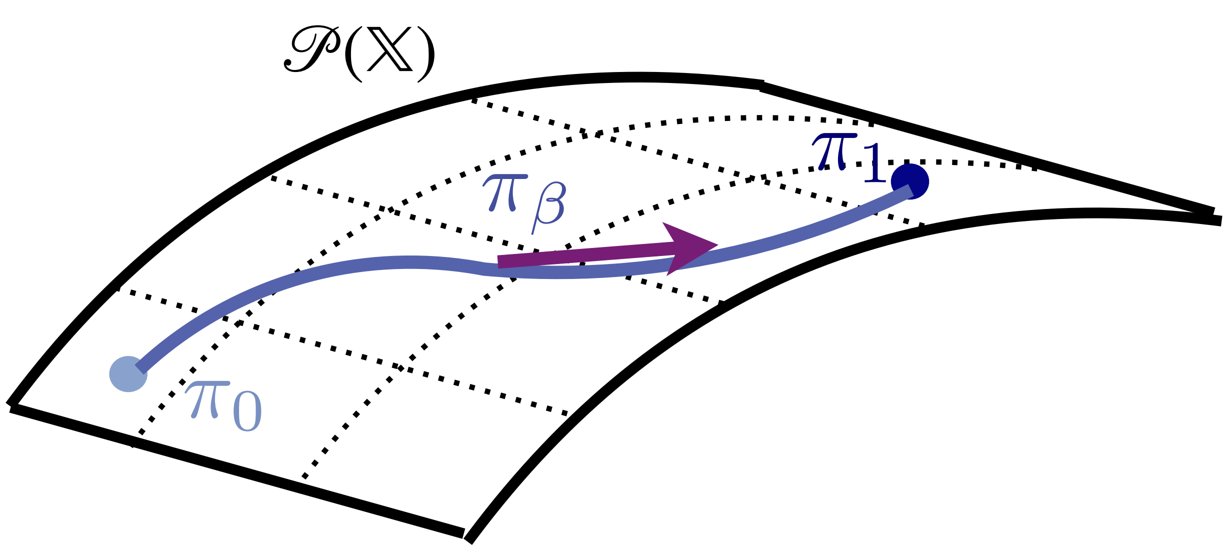


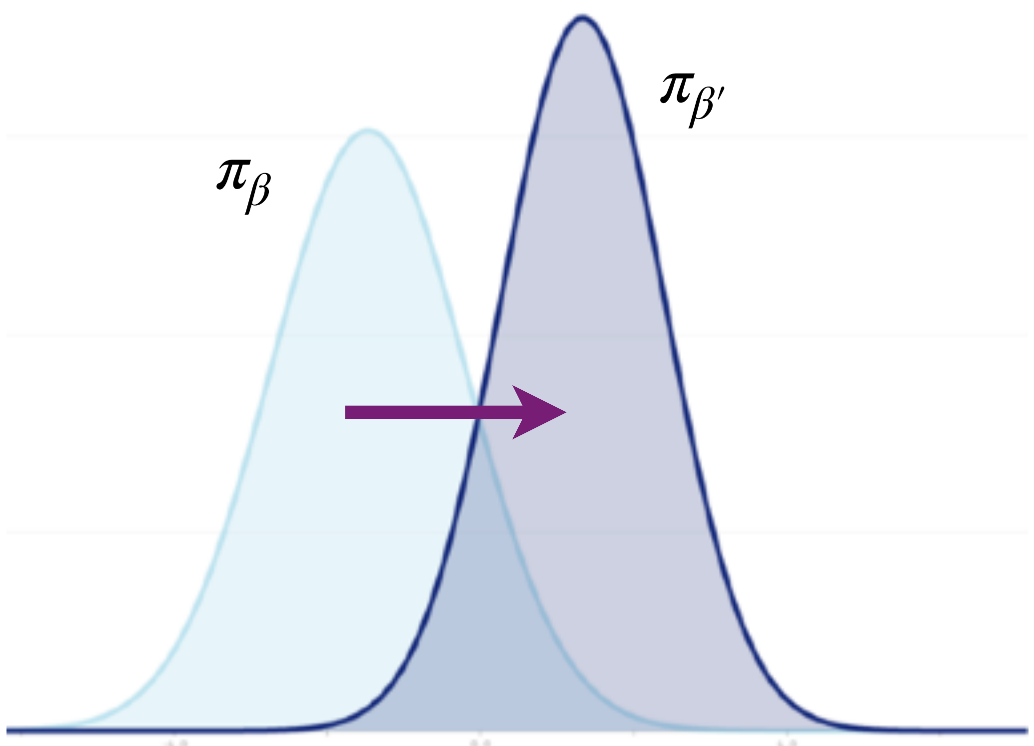
- We can measure how much the distribution changes from β to β' through the likelihood ratio

ANNEALING AS A CURVE

- Suppose $\mathscr{P}(\mathbb{X})$ is the space of probability densities supported on $\mathbb{X}$
- $\beta \mapsto \pi_\beta$ defines a curve in $\mathscr{P}(\mathbb{X})$ parametrised by β with **position** π_β
- We can measure how much the distribution changes from β to β' through the likelihood ratio
- The change in position $\Delta\pi_{\beta,\beta'} : \mathbb{X} \mapsto \mathbb{R}_+$ as the percentage change in likelihood

$$\Delta\pi_{\beta,\beta'}(x) = \frac{\mathrm{d}\pi_{\beta'}}{\mathrm{d}\pi_\beta}(x) - 1 = \frac{\pi_{\beta'}(x) - \pi_\beta(x)}{\pi_\beta(x)}$$





VELOCITY OF DENSITY